



# ROBOGAMBIT

*Smart Moves. Precision Play.*

AI Chess Robot • Computer Vision • ROS 2 • PySide6 GUI

Intelligent Robotics Course — Final Project

- **What is RoboGambit?**

## **A robot that plays chess on a real board.**

A camera watches the game. Computer vision detects your move. Stockfish thinks. A 5-DOF robotic arm physically picks up and places the piece — while a polished GUI mirrors everything in real time.



### **SEE**

Camera + OpenCV



### **THINK**

Stockfish AI



### **MOVE**

Robotic Arm

# ● System Architecture

*Modular pipeline. Loosely coupled. ROS 2 between layers.*



Communication via ROS 2 topics: `/robogambit/move` → `/nano_serial`

- 🔧 Each layer can be swapped or tested independently
- 🔄 Standard ROS topics — easy to extend with feedback loops
- 🖥️ Same GUI works in fake-mode (no hardware) or real-mode

# ● Computer Vision — Detecting Moves on a Real Board



64-square pixel calibration



## Board Calibration

Click 4 corners → maps every chess square to a pixel polygon (sqdict.json).



## Frame Differencing

Compare before/after frames → finds the 2 changed squares = your move.



## Auto-Detect Mode

Motion-then-stillness state machine: just play, no buttons to click.



## Live Preview

Camera feed embedded in the GUI — see exactly what the system sees.

# ● AI Engine & Game Modes



## Stockfish Engine

- World-class chess engine integrated via python-chess
- Three difficulty levels: Easy / Medium / Hard
- Skill levels mapped to Stockfish skill setting
- Smart move hints — toggleable per player
- Full chess rule support: captures, castling, en passant, promotion
- Move legality validation — invalid moves are gracefully rejected

## Four Game Modes



### Player vs AI (Robot)

Real board, real arm



### Player vs AI

On-screen vs Stockfish



### Player vs Player

Two humans + hints

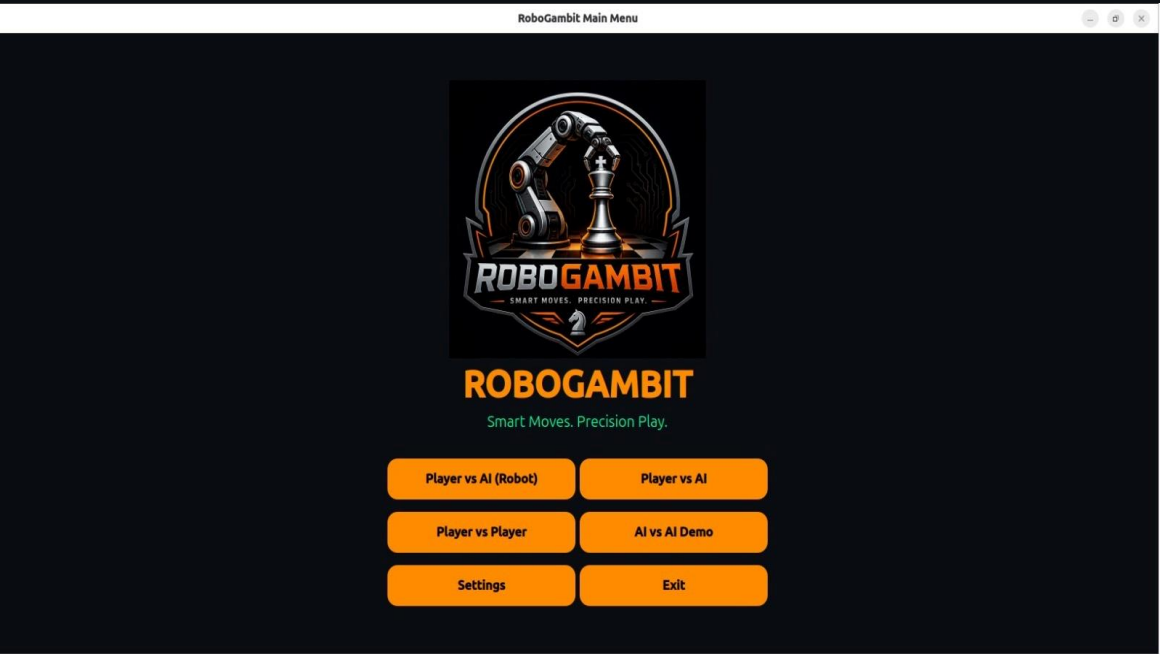


### AI vs AI Demo

Watch Stockfish self-play

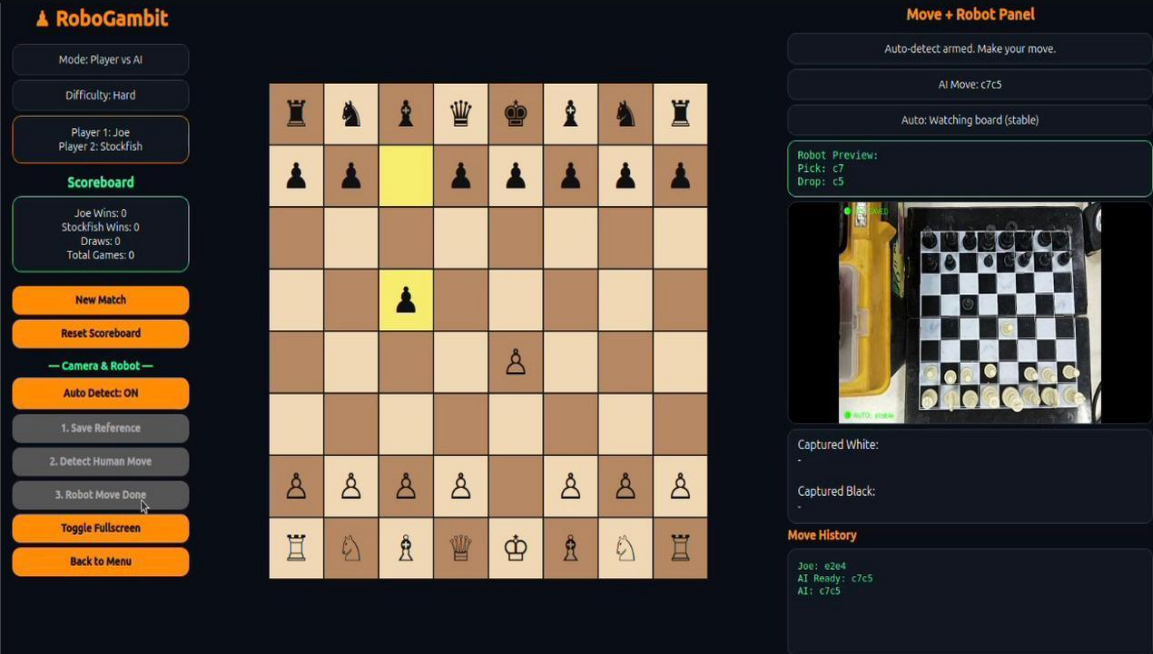
# ● GUI Showcase

*Built with PySide6 — dark theme, fullscreen, and live camera feed.*



**Main Menu**

*Game-mode selection with branded logo*



**In-Game View**

*Board + live camera + history + robot status*

🎯 Capture highlighting

📜 Move history

🏆 Scoreboard

💡 Per-player hints

🖥️ Fullscreen mode

# ● Hardware & Arduino Firmware



## The Arm — 5 DOF

- 5 servos: base, joint 1-3, gripper
- Arduino Nano controller (115200 baud serial)
- External 5V supply for servo power rail
- Pick-and-place pattern: hover → pick → grip → travel → place → release
- Off-board graveyard zones for captured pieces
- Spare-piece slots for promotion handling

## chess\_arm\_controller.ino

```
#include <Servo.h>
Servo s1, s2, s3, s4, s5;

void setup() {
  Serial.begin(115200);
  s1.attach(3); s2.attach(5);
  s3.attach(6); s4.attach(9);
  s5.attach(10); // gripper
}

void loop() {
  // Read "a1,a2,a3,a4,a5\n"
  // Drive 5 servos to angles
}
```

# • ROS 2 Control Pipeline

*From chess move to servo angles — three nodes, two topics, one robot.*

1

## GUI Publisher

Publishes UCI move strings

`main.py → robot/ros_robot.py`



`/robogambit/move → "e2e4"`

2

## IK Translator Node

Chess square → 5 servo angles

`Calibrated lookup table per square`



`/nano_serial → "90,45,120,60,30"`

3

## Serial Bridge Node

Forwards angles to Arduino over USB

`Auto-detects Arduino by VID/PID`



# ● The Team

*Intelligent Robotics Course — Final Project*



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*GUI • Vision • AI Integration*



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*Mechanical • Robotic Arm • Wiring*

SEE. THINK. MOVE.

*Live demo coming up next.*

 RoboGambit